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James Clements III

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Summary

Senior audio DSP engineer with seven years of shipped real-time C++ on embedded ARM and desktop platforms. Particular focus on convolution reverberation, room correction, and low-latency scheduling. Comfortable across the full audio stack from kernel-level driver glue to plug-in UI in JUCE.

Education

Stanford University — M.S., Music, Science & Technology (CCRMA), 2019. Thesis: *Sparse Time-Varying Filters for Physically-Modelled Strings*.

Oberlin Conservatory + College — B.A. Computer Science, B.Mus. Harp Performance (dual-degree program), 2016. *Magna cum laude*.

Professional experience

Cantus Audio, Portland OR — *Lead audio engineer*, 2022 – present.

- Own the real-time engine in *Cantus Verb Pro*: a 96 kHz partitioned convolution reverb shipping on macOS, Windows, and an in-house ARM hardware unit (Cortex-A78).
- Brought worst-case 64-sample block latency from 11 ms to 2.8 ms by redesigning the FFT block scheduler around lock-free SPSC queues.
- Mentor three junior engineers and run the team's weekly DSP reading group.

Center for Computer Research in Music and Acoustics, Stanford CA — *Research engineer*, 2019 – 2022.

- Core contributor to the *faust2* compiler; landed the SSA-based delay-line optimiser used in versions 2.50 onward.
- Co-authored two ICASSP papers (see Publications) on physically modelled string synthesis.

- Built and maintained the lab's nightly regression suite (1300 tests, ~9 minutes wall-time on a 32-core machine).

Smule, San Francisco CA — *Audio engineer intern*, summer 2018.

- Implemented the pitch-correction path for *Smule Sing!* on iOS, using a phase-vocoder front end and an autocorrelation-based pitch tracker.

Selected publications & projects

- Clements, J. "Sparse, Time-Varying All-Pole Models for Plucked Strings," ICASSP 2022. DOI: 10.1109/ICASSP.2022.9747033.
- Clements, J. and Smith III, J. O. "A Lazy Scheduler for Faust Block Diagrams," ICASSP 2021.
- **abcjsharp** — fork of `abcjs` with proper harp grand-staff rendering; ~6k LOC of TypeScript. github.com/clements/abcjsharp.
- **mdlout** — Markdown to PDF (via Lout) and HTML (via SVG) pipeline. The document you are reading was built with it.

Technical skills

Languages C++ 17 (daily), Python 3 (daily), TypeScript, Rust, Faust.

DSP Convolution, FFT-based filtering, phase vocoders, FDN reverb, physical modelling, room correction, psychoacoustic masking.

Audio I/O CoreAudio, ASIO, ALSA, VST3, AU, AAX, WebAudio.

Tooling Make, Bazel, perf, Tracy, ASan, TSan, UBSan, Ableton Live, Pro Tools.

Misc Concert-level pedal harp; conversational French; amateur radio (KC7QHK).

References

Available on request.